MIL-I-27304E(USAF)
4 April 1977
SUPERSEDING
MIL-I-27304D(USAF)
14 February 1974

MILITARY SPECIFICATION

INDICATOR, RATE OF FLOW

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for types of hermetically sealed indicators for indicating the flow rate of engine fuel in pounds per hour (pph). The indicator is a remote second-harmonic selsyn receiver.

* 1.2 Classification. The indicators shall be of the following types, as specified (see 6.2). The Type EFU-15A/A series indicators contains a low frequency solid state power supply to drive one or two Type TRU-31A/A rates of flow transmitters conforming to MIL-T-27305. The Type EFU-42/A series indicators are repeaters only and do not contain the power supply. (See 6.1).

EFU-15A/A-4	Integrally lighted (red)	0 - 1,200 pph
EFU-15A/A-6	Integrally lighted (red)	0 - 2,500 pph
EFU-15A/A-7	Integrally lighted (white)	0 - 1,200 pph
EFU-15A/A-10	Nonintegrally lighted	0 - 800 pph
EFU-15A/A-12	Integrally lighted (white)	0 - 800 pph
EFU-15A/A-18	Integrally lighted (white)	0 - 5,000 pph
EFU-42/A-18	Integrally lighted (white)	0 - 5,000 pph

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-636	Box, Fiberboard

FSC 6620

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: ASD/ENESS, Wright-Patterson AFB, OH 45433 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-116	Preservation-Packaging, Methods of
MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-C-14806	Coating, Reflection Reducing, for Instrument Cover Glasses and Lighting Wedges
MIL-L-25142	Luminescent Material, Fluorescent
MIL-L-25467	Lighting, Integral, Aircraft Instruments, General Specification for
MIL-C-26500	Connectors, General Purpose, Electrical, Miniature, Circular, Environment Resisting, General Specification for
MIL-L-27160	Lighting, Instrument, Integral, White, General Specification for
MIL-T-27305	Transmitter, Rate of Flow TRU-31A/A, Fuel
MIL-B-27497	Bearings, Jewel, Sapphire or Ruby, Synthetic

STANDARDS

FEDERAL

FED-STD-595	Colors

MILITARY

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-701	Lists of Standard Semiconductor Devices
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MIL-STD-889	Dissimilar Metals
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, with Appropriate Test Methods
MS24266	Connectors, Plug, Electrical-Straight, Miniature
MS27034	Connector, Receptacle, Electrical Pin Insert, Cylindrical, Miniature Hermetic, Solder Mount
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS33585	Pointers, Dial, Standard Design of Aircraft Instrument
MS33639	Cases, Instrument, Clamp-Mounted, Aircraft

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification. The indicators furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening for bids (see 4.4 and 6.3).

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3.2 Standards and specifications. Standards and specifications for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Preliminary list of electronic parts. A preliminary list of electronic parts shall be furnished directly to the procuring activity in accordance with MIL-E-5400 prior to ordering or fabricating parts for the production quantity. The parts shall conform to MIL-E-5400. Nonstandard parts shall be approved by the procuring activity prior to the fabrication of the completed item. No changes shall be made in the approved nonstandard parts. Whenever practicable, standard electronic parts shall be used.

3.4 Materials

3.4.1 Nonferrous materials. Nonferrous materials shall be used for all parts of the indicator, except where ferrous materials are essential.

3.4.2 Toxic and corrosive fumes. The indicator shall be of such material and construction that when the material is burned, fumes or gases that combine with the atmosphere to form acid or corrosive alkali will not be liberated.

3.4.3 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the indicator except where magnetic materials are essential.

3.4.4 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service.

3.4.4.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.4.5 Protective treatment. When materials are used in the construction of the indicator that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the requirements of this specification. The use of protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.5 Design and construction

3.5.1 Function. Both the EFU-15A/A and EFU-42/A series indicators shall include all of the circuitry and mechanisms necessary to convert a second harmonic signal from the TRU-31A/A transmitter conforming to MIL-T-27305, a pointer indication of flow rate in pounds per hour of aviation fuel. The type 15A/A indicator shall also supply power to the transmitter to drive the impeller motor at a constant speed. The indicator and transmitter shall operate as a system. The power supply shall be solid state and shall operate satisfactorily and meet the requirements herein when the output circuit of a de-energized power supply is connected to it in parallel with the transmitter impeller motor load.

3.5.1.1 Operation. The indicator shall be designed to operate on the second harmonic system principle in which the moving element consists of nonenergized permanent 2-pole magnet or magnets and the torodially wound stator is energized with 115V, 400-Hz, single-phase a-c power. Remote indication of the angular position of the turbine in the TRU-31A/A transmitter is obtained through the

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use of the coil and core operating on the permanent magnet rotor. The indicator rotor repeats the position of the transmitter rotor (turbine). The power supply in the type EFU-15A/A indicator provides a constant signal to the transmitter impeller motor so that the transmitter impeller is driven at a constant speed. The direct-current resistance of the pickoff coil sections shall be between 290 and 350 ohms per section when the coil temperature is 25° ±5°C.

3.5.2 Input signals

* 3.5.2.1 Rate of flow transmitter. The signal from the transmitter shall be of the second harmonic type. The TRU-31A/A transmitter shall be so calibrated that the turbine will rotate approximately 307° for a full-scale reading. For an increasing flow rate, the turbine shall rotate in a clockwise direction when viewed from the inlet end. The full-scale range of the transmitter is as follows:

TRU-31A/A-1	0 - 1,200
TRU-31A/A-3	0 - 2,500
TRU-31A/A-4	0 - 800
TRU-31A/A-6 & 8	0 - 5,000

* 3.5.2.2 Power. The indicator shall be designed to operate from 115V ±10 percent 400 Hz ±10 percent single-phase a-c power in accordance with the characteristics of MIL-STD-704. In addition, the indicator shall operate from power having a frequency variation as great as ±20 percent except that the ratio of frequency to voltage shall not deviate from the normal (400/115) by more than ±22 percent.

3.5.3 Output signals

* 3.5.3.1 Power supply. The EFU-15A/A indicators shall supply 34 ±4V rms 6 Hz ±0.05 percent 2-phase power when operating either one or two TRU-31A/A transmitters conforming to MIL-T-27305.

3.5.3.2 Power supply warmup. After a warmup period of 1 minute, the power supply shall be within 6 or 12 Hz ±0.09 percent and in conformance with table I, as applicable, over an operating temperature of -54° to +71°C.

3.5.3.2.1 The power supply shall be designed to meet the accuracy limits specified in table II within the time periods and at the temperatures specified.

3.5.4 Indicator warmup. The indicator, exclusive of the power supply, shall not require a warmup period exceeding 10 seconds to meet the requirements specified herein.

3.5.5 Reliability. The indicator shall have a minimum acceptable mean-time-between-failures (MTBF) of 1,200 hours demonstrated to a 90 percent confidence level.

3.5.5.1 Longevity. The indicator shall have an operating life span (equipment longevity as defined in MIL-STD-781) of not less than 2,500 hours before wearout failures occur or the equipment consistently fails to meet the specified MTBF index. Parts requiring replacement within this period of time and the normal operating period of such parts shall be reported by the contractor to the procuring activity.

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* TABLE I. Indicator tolerances.

Requirements	±PPH Tolerances			
	0-800 Range	0-1,200 Range	0-2,500 Range	0-5,000 Range
Position	2	4	8	16
Hysteresis	2	4	8	16
Vibration				
Pointer oscillation (double amplitude)	3	6	12	24
Pointer variation	2	4	8	16
Friction				
Room and high temperature	2	4	8	16
Low temperature	3	6	12	24
Voltage and frequency variation				
Room temperature	3	5	10	21
Power supply	Tolerance (Percent of Rated Frequency)			
Room temperature	±0.05	±0.05	±0.05	±0.05
Voltage and frequency error	±0.07	±0.07	±0.07	±0.07
High temperature	±0.09	±0.09	±0.09	±0.09
Low temperature	±0.09	±0.09	±0.09	±0.09
Vibration resonance	±0.09	±0.09	±0.09	±0.09
Room temperature following reliability	±0.10	±0.10	±0.10	±0.10

TABLE II. Accuracy Limits.

AMBIENT TEMPERATURE	WARMUP PERIOD	OUTPUT FREQUENCY
a. -54°C	10 seconds	6 Hz ±0.1 percent or 12 Hz ±0.1
b. +25°C	10 seconds	6 Hz ±0.1 percent or 12 Hz ±0.1
c. +71°C	10 seconds	6 Hz ±0.1 percent or 12 Hz ±0.1

3.5.6 Maintainability. The indicator shall be so designed as to provide ease of maintenance which shall include the following:

a. Minimization of the complexity of maintenance tasks (for example: calibration, adjustments, inspection, et cetera) by maximum use of simple design which includes optimum interchangeability and use of standardized equipment or commercial items.

b. Rapid and positive recognition of equipment malfunction or marginal performance.

c. Rapid and positive identification of the replaceable defective part, assembly or component.

d. Minimum numbers and types of tests and test equipment (special and standard) needed to perform maintenance.

e. Optimum accessibility in all components requiring maintenance inspections, removals, or replacements.

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3.5.7 System compatibility. The EFU-15A/A and EFU-42/A indicators shall be compatible with the TRU-31A/A transmitters for the corresponding ranges and shall be interchangeable with previously supplied indicators of the same type.

3.6 Performance. The indicator shall be capable of meeting the requirements specified herein when subjected to the following conditions:

a. Temperatures - Operating temperatures ranging from -54° to +71°C and storage temperatures from -62° to +80°C.

b. Vibration - Vibration of 2g for 9 hours at a double amplitude of 0.00016 to 0.100 inch and at 5 to 500 Hz, as specified.

c. Salt fog - Exposure to salt sea atmosphere for a 48-hour period.

d. Thermal shock and fogging - Alternate immersion for 30 minutes in water at 85°C and then at 5°C for 8 cycles as specified in section 4.

e. Altitude - Absolute pressure equivalent to 100,000 feet.

f. Voltage and frequency variation - Voltage and frequency variations ranging from 104V 320 Hz to 126V 480 Hz.

g. Shock - Up to 15g.

3.6.1 Electromagnetic interference. The indicator shall meet the electromagnetic interference requirements of MIL-STD-461 for class ID equipment.

3.6.2 Scale error. Scale error shall be within the tolerances specified in tables I and III. The scale error includes the tolerances of the test equipment.

3.6.3 Magnetic effect. The indicator, operating or nonoperating, shall not create a magnetic field with an intensity that will deflect a short bar magnetic compass more than 3°.

3.6.4 Response. The time required for the indicator to change from zero to half scale or from half scale to zero shall not exceed 3 seconds under room temperature and high temperature conditions when tested in accordance with 4.6.5.

3.6.5 Power consumption. The power consumption of the indicator (including the power supply) shall not exceed 30 va with the transmitter load connected but exclusive of the lighting circuit. The 5V a-c or d-c power consumption in the lighting circuit shall not exceed 1 va/bulb.

3.6.6 Damping. The indicator shall be capable of attenuating small oscillatory transmitter output signals. The indicator damping shall minimize indicator response to fluctuations in fuel of ±1.5 percent of full scale at a frequency of 2.0 ±0.5 Hz.

3.6.7 Dielectric strength. The indicator shall withstand applications of 250V at commercial frequency applied between each pin not connected to ground, and the case for a period of 5 seconds after the indicator case is sealed.

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TABLE III. Indicator fuel flow rate tolerance ±PPH.

Fuel Flow	0-800 Range		0-1200 Range		0-2500 Range		0-5000 Range	
Test Points PPH	Rm & High Temp	Low Temp & Reliab	Rm & High Temp	Low Temp & Reliab	Rm & High Temp	Low Temp & Reliab	Rm & High Temp	Low Temp & Reliab
0	4	6	6	10	11	18	24	37
100	3	5	5	8				
150	3	5						
200	3	5	5	8	10	15	22	34
250	3	5						
300	3	5	5	8				
350	3	5						
400	3	5	5	8	10	15	20	32
450	3	5	4	7				
500	3	5	4	7				
600	3	5	4	7	10	15	20	32
700	3	5	5	8	10	15		
800	3	5	5	8	10	15		
900			5	8	8	13		
1,000			5	8	8	13	20	32
1,100			5	8	8	13		
1,200			5	8	10	15		
1,400					10	15	17	28
1,600								
1,800					10	15	17	28
2,000								
2,200					10	15	17	28
2,400								
2,500					10	15	17	28
2,600								
2,800								
3,000							20	32
3,500							20	32
4,000							20	32
4,500							22	34
5,000							22	34

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3.6.8 Fogging. There shall be no evidence of moisture or oil fogging of the cover glass when the indicator is subjected to a temperature of 85° ±2°C for a minimum of 1 hour and then the face of the glass immediately immersed in a 5°C bath for a period of 1 minute.

3.6.9 Hysteresis. Hysteresis shall not exceed the tolerances specified in table I.

3.7 Case. The case shall conform to MS33639 for the 2-inch size, shall be finished in durable black, Color No. 37038 in accordance with FED-STD-595, and shall be hermetically sealed. The front edge of the case shall not extend beyond the outer surface of the cover glass by more than 0.032 inch. The case length for the lighted EFU-15A/A indicators shall not exceed 6.375 inches exclusive of the electrical receptacle. The case length for the EFU-42/A indicators shall not exceed 3.50 inches exclusive of the electrical receptacle.

3.7.1 Filling medium. The filling medium shall be of at least 98 percent purity, free of dust particles, and shall contain no more than 0.006 milligram of water vapor per liter (dewpoint -65°C) at the filling pressure. The filling medium shall be 88 to 92 percent nitrogen and the remainder helium. The absolute pressure of the filling medium in the case shall be 12 to 14.7 psi.

3.7.2 Filler tube. A tube or other facility for evacuating and filling the case shall be provided at the rear of the case and shall be recessed beneath the level of the backplate surface.

3.8 Dial. The indicator dial shall conform to figure 1. The style and proportions of numerals and letters on the dial shall conform to MS33558 except as specified herein. The dial shall have a minimum diameter of 1.75 inches as measured at the outside ends of the scale graduations and shall be finished in black, Color No. 37038, in accordance with FED-STD-595. The calibration of fuel flow versus angular degrees shall be linear within ±0° 15' of arc within the 0 to full scale flow range.

3.8.1 Dial markings. The dial markings for nonlighted indicators shall be fluorescent-luminescent in accordance with MIL-L-25142. For integrally lighted indicators, the dial markings shall be finished in white, Color No. 37875 in accordance with FED-STD-595. The dimensions shall be as specified in table IV.

* 3.9 Lighting. Internal lighting shall be designed into the indicator. The design of the indicator shall be such that the internal lighting may be either white or red depending upon the color specified. With the exceptions specified herein, the red lighting shall be in accordance with MIL-L-25467 and the white lighting shall be in accordance with MIL-L-27160. The internal lighting of the prime standard and all qualification indicators shall be subject to the approval of the qualifying activity.

* 3.9.1 Lamps. Lamps shall be in accordance with MIL-L-25467 or MIL-L-27160.

* 3.9.2 Prime standard. The internal lighting shall be comparable to and shall balance in brightness with a prime standard indicator. The prime standard shall be either an actual indicator conforming to the requirements of this specification or a lighted model. The prime standard shall be used as the reference for acceptance of all production indicators.

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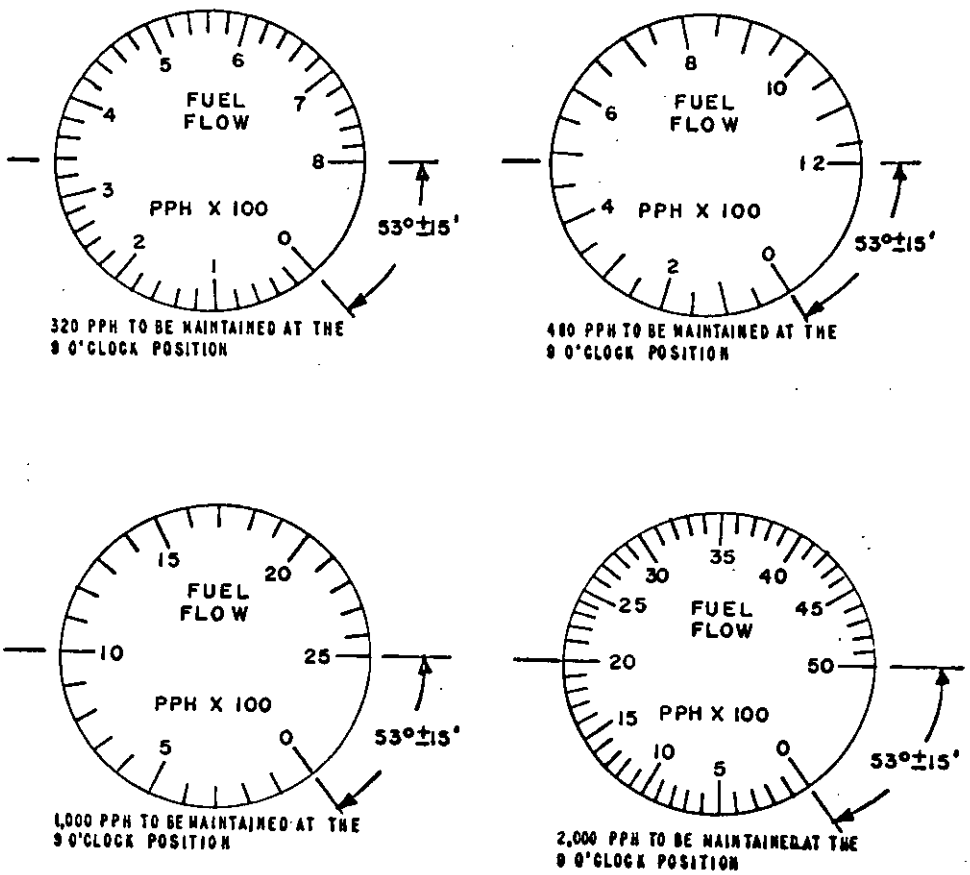


FIGURE 1. Dial layout.

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TABLE IV. Dial marking dimensions.

	HEIGHT OR LENGTH INCH ±0.016	WIDTH OF LINE OR GRADUATION INCH ±0.005
Scale numerals 0 through full scale	0.156	0.031
100 pph graduation on 800 pph dial	0.156	0.031
100 pph graduation on 1,200 pph dial	0.156	0.031
0 and 500 pph graduation on 2,500 pph dial	0.156	0.031
0 and 500 pph graduation on 5,000 pph dial	0.156	0.031
20 pph graduation on 800 pph dial	0.125	0.020
50 pph graduation on 1,200 pph dial	0.125	0.020
100 pph graduation on 2,500 pph dial	0.125	0.020
100 pph graduation on 5,000 pph dial	0.125	0.020
Lettering PPH x 100	0.063	0.010
Lettering FUEL FLOW	0.094	0.016

* 3.9.2.1 Prime standard recertification. All representative areas of the face of the prime standard indicator shall be measured for brightness in foot-lamberts. Each area shall be measured at a sufficient number of locations to obtain data representative of each area including the brightest and dimmest portions of the indicator in each display. The prime standard indicator shall be measured periodically (at least every 50 hours of operation or every 120 days, whichever occurs first) to determine any brightness or color change. The data and, if required by the qualifying activity, the prime standard shall be submitted for evaluation and reapproval or corrective action, or both.

* 3.9.2.2 Visual comparison. The lighting level of the indicator shall fall within the limits of the prime standard when energized at the voltage levels specified in 4.6.24.2.2.

3.10 Pointer. The pointer shall conform to MS33585-8 and shall be suitable for integral or nonintegral lighting, as applicable.

3.11 Cover glass. All reflecting glass surfaces shall be provided with a reflection reducing coating which meets the requirements of MIL-C-14806 and the environmental conditions specified herein.

3.11.1 Dial to cover glass distance. The distance between the inner surface of the cover glass and the indicator dial shall be the minimum practicable and shall not exceed 0.188 inch. The major and one-quarter of the minor graduations shall be visible when viewed from any direction within a 25° angle from a line perpendicular to the center of the indicator dial.

3.12 Synthetic jewel bearings. When synthetic sapphire or ruby jewel bearings are used, they shall be in accordance with MIL-B-27497.

3.13 Screw threads. Unless otherwise specified, the threads of all machine screws shall conform to MIL-S-7742.

* 3.14 Electrical receptacle. The EFU-15A/A-4, -6, -7, -10, and -12 indicators shall be provided with an MS27034H12B12P hermetically sealed receptacle conforming to MIL-C-26500. The EFU-15A/A and EFU-42/A-18 indicators shall be provided with an MC14H-13-12P cannon receptacle or equivalent. Wiring for the

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EFU-15A/A-4, -6, -7, -10, and -12 indicators is shown on figure 2 and for the EFU-15A/A and EFU-42/A-18 indicators as shown on figure 3. Two EFU-15A/A-18 indicators may be connected together as shown in figure 4.

3.15 Transistors and diodes. Transistors and diodes used in the indicator shall be of silicon type designed for continuous operation in a 70°C ambient temperature in accordance with MIL-STD-701. Circuit design shall be such that no electronic component will be operated outside its rated characteristics with 115V ±10 percent 400 Hz ±20 percent power input and with indicator input signals as specified herein. No electronic device shall break down when excited by transients as specified in MIL-STD-704.

3.16 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part number and changes thereto.

3.17 Weight. The weight of the EFU-15A/A indicator shall not exceed 1.35 pounds and the weight of the EFU-42/A indicator shall not exceed 0.8 pounds.

3.18 Electrical schematic. An electrical schematic showing the function of each connector pin shall be permanently attached to the indicator. A metal decalcomania may be used provided it is permanently attached to the case and can not be peeled back.

3.19 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.19.1 Nameplate. A nameplate shall be provided on the rear of the case. When applicable, the information shall conform to MIL-STD-130. Metal decalcomanias shall not be used.

3.20 Workmanship. The indicator, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, brazing, painting, riveting, machine-screw assemblies, and freedom of parts from burrs and sharp edges.

3.20.1 Dimensions and tolerances. Dimensions and tolerances not specified shall be as close as is consistant with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation or performance of the indicator, they shall be held or limited accordingly.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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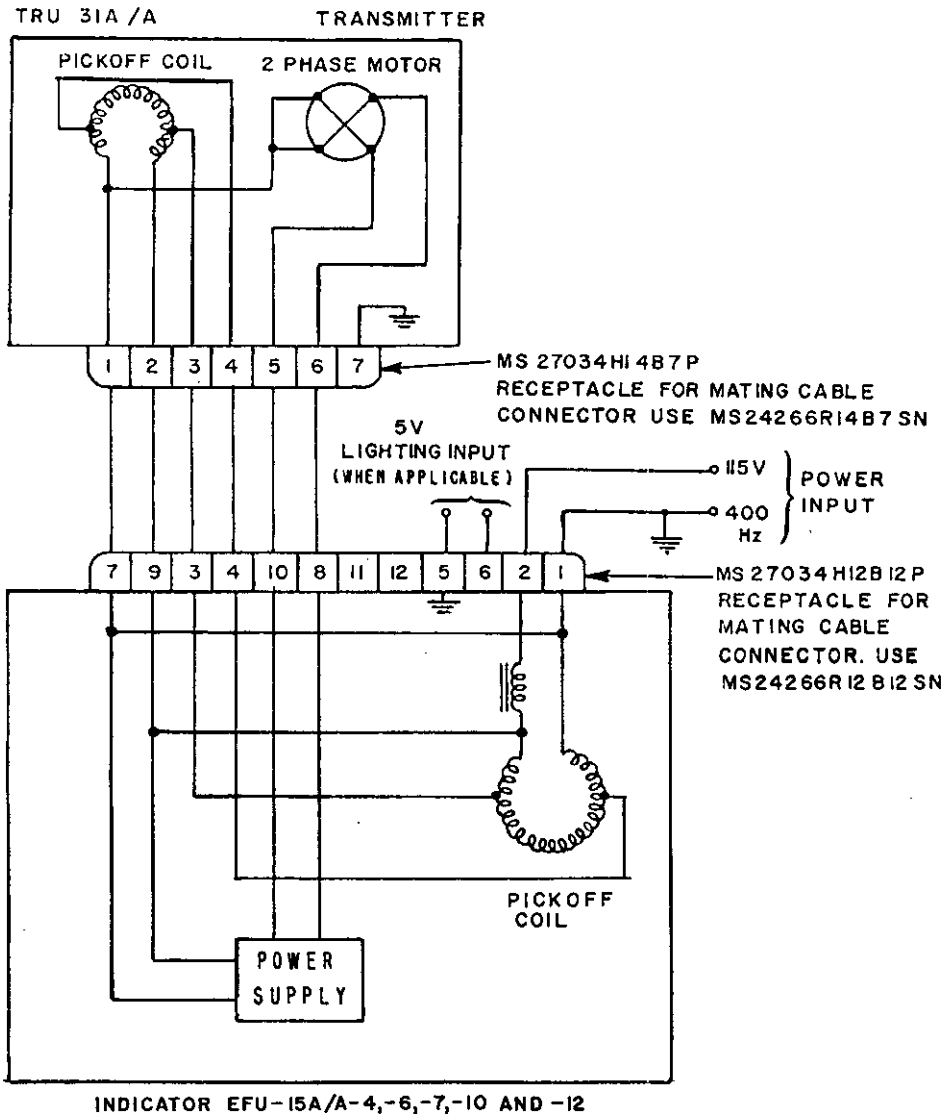


FIGURE 2. System connection diagram.

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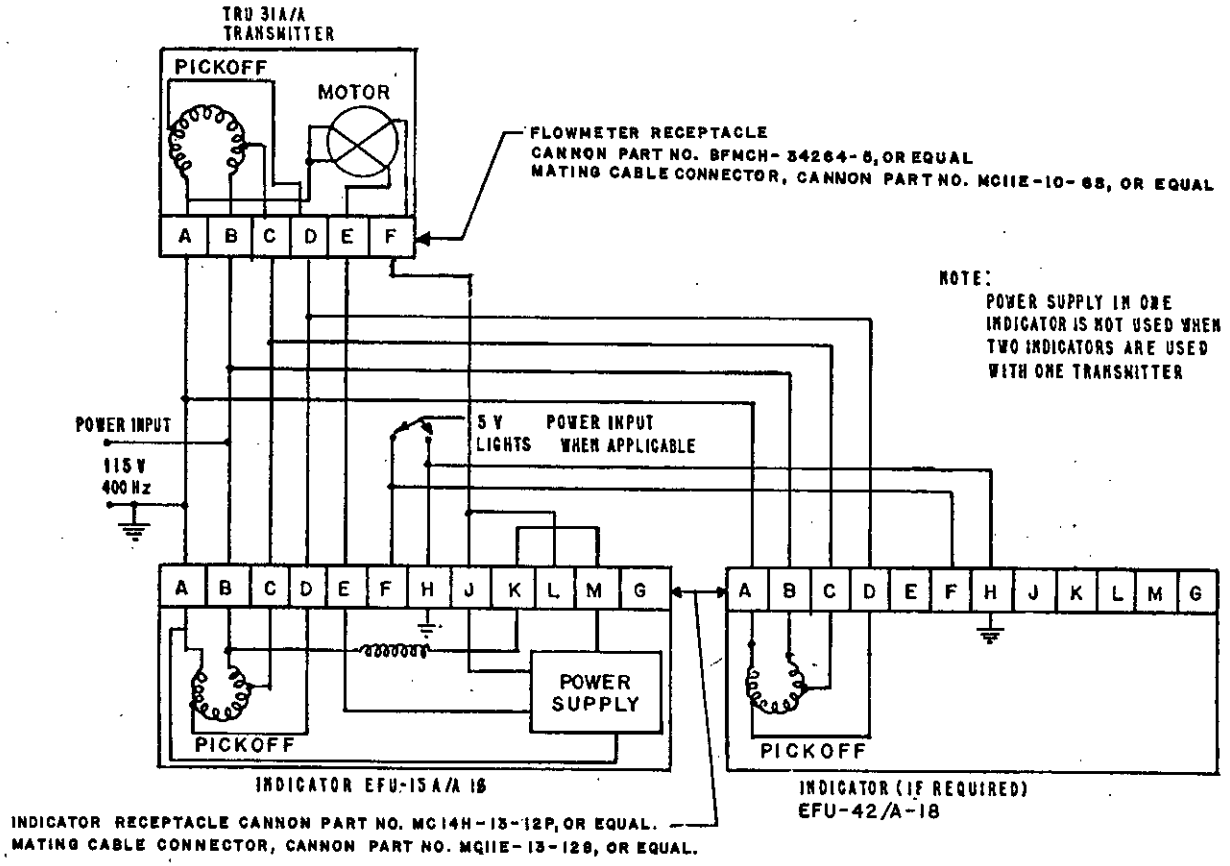


FIGURE 3. System connection diagram.

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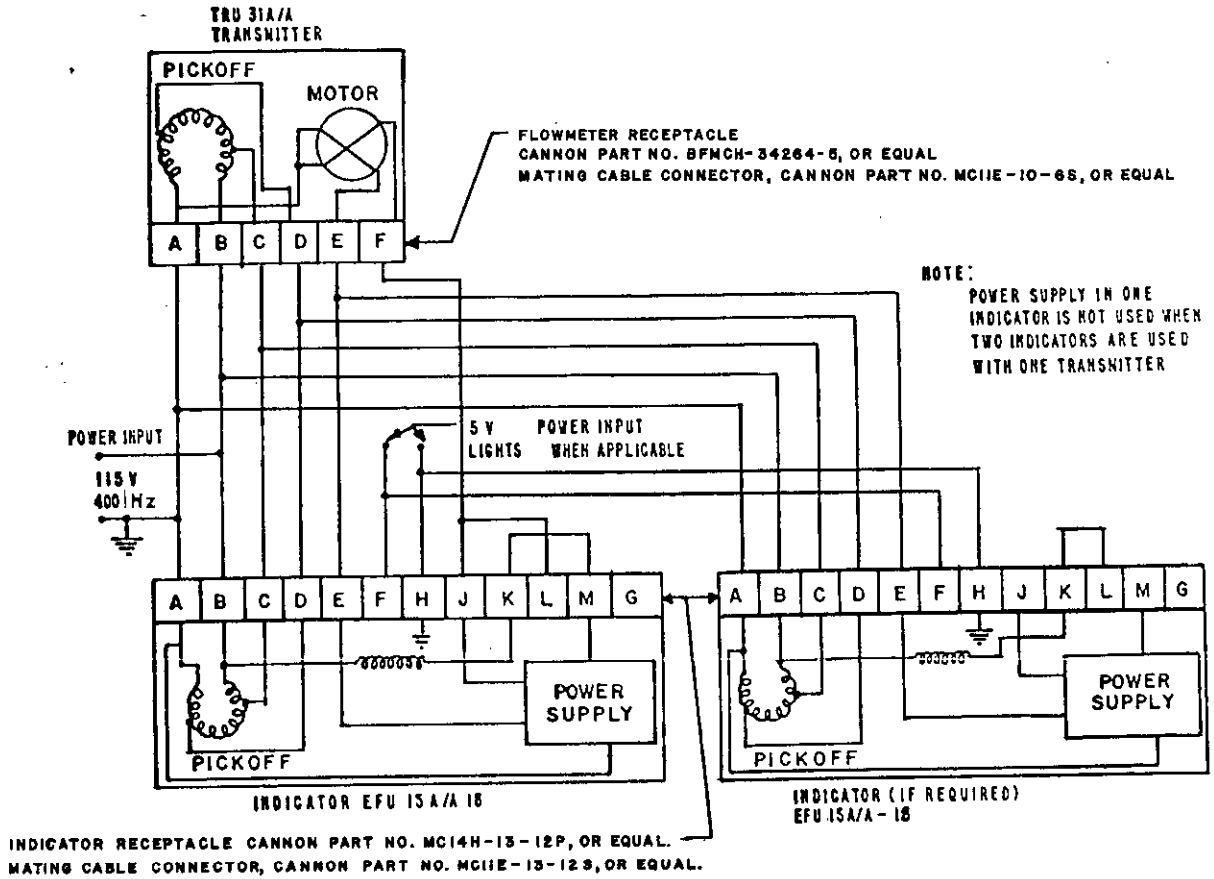


FIGURE 4. System connection diagram for EFU-15A/A-18 indicator.

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4.2 Classification of tests. The inspection and testing of indicators shall be classified as follows:

a. Qualification of tests.

b. Quality conformance tests.

4.3 Test conditions

4.3.1 Vibration. Unless otherwise specified, the indicator may be gently vibrated before a test reading is taken.

4.3.2 Vibration stand. Whenever a vibration stand is specified, the stand shall vibrate at any desired frequency between 5 and 500 Hz.

4.3.3 Test positions. Unless otherwise specified, the indicator shall be tested with the case in a horizontal position with the scale upright.

4.3.4 Master transmitter. The indicator shall be tested with a master transmitter that generates a signal in accordance with 3.5.2.1. Sufficient tests shall be performed to determine that the transmitter signal is not in error in excess of 0.01 percent.

4.3.5 Tolerances. Test tolerances shall be as specified in tables I and III.

4.3.6 Test power. Unless otherwise specified, the indicator shall be tested with 115 ±2V, 400 ±2 Hz single-phase, a-c power, and 5.0 ±0.1V a-c or d-c power for lighting.

4.3.7 Test load. Unless otherwise specified, the EFU-15A/A series indicators shall be tested while connected to the transmitter or simulated transmitter load. The test wiring shall be in accordance with figure 5. The EFU-42/A series indicators shall be tested using the master transmitter only.

4.4 Qualification testing

4.4.1 Test samples. The test samples shall consist of at least 14 indicators representative of the production equipment. The samples shall be identified with the manufacturer's part number and such other information as required by the qualifying activity. Three of the samples shall be subjected to all the tests except reliability and longevity, 11 samples shall be subjected to the individual and reliability tests and 3 of the 11 samples shall be subjected to the longevity test.

4.4.2 Qualification tests. The qualification tests shall consist of all the tests specified under 4.6.

4.5 Quality conformance tests. The quality conformance tests shall consist of:

a. Individual tests

b. Sampling tests.

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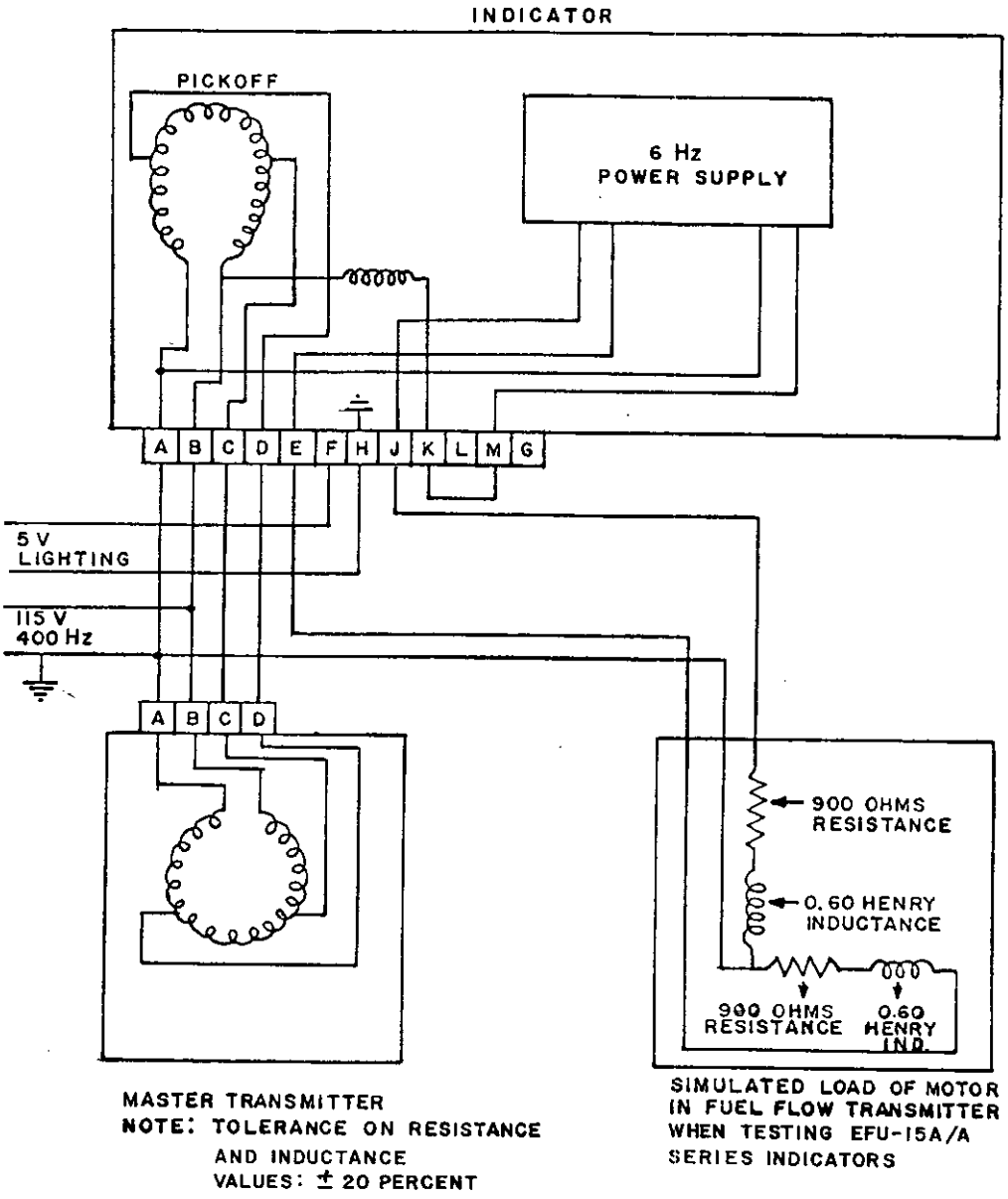


FIGURE 5. Indicator test wiring.

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4.5.1 Individual tests. Each indicator shall be subjected to the following test as described under 4.6:

a. Examination of product

b. Dielectric strength

c. Scale error and friction

d. Hysteresis

e. Response

f. Sealing

g. Lighting (visual comparison)

h. Damping

i. Fogging.

4.5.2 Sampling tests. A minimum of three indicators shall be selected at random from each 100 or fraction thereof produced on the contract or order and subjected to the following tests as described under 4.6:

a. Individual

b. Voltage and frequency variation

c. Frequency cycling

d. Voltage surge

e. Vibration error

f. Low temperature

g. High temperature

h. High altitude-low temperature

i. Position error

j. Thermal shock and fogging

k. Power consumption

l. Lighting (sampling plan A test of MIL-L-25467 or MIL-L-27160).

4.5.2.1 Reliability. For reliability sampling, a lot shall consist of all indicators produced on a specific contract. A minimum of five indicators shall be selected at random from each 100 or fraction thereof produced on the contract and subjected to the test specified in 4.6.26.2.

4.5.2.2 Rejection and retest. When an indicator selected from a production run fails to meet the specification, no indicators still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain to the

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Government representative the cause of failure and the action taken to preclude recurrences. After the corrections have been made, such tests as determined by the qualifying activity shall be repeated.

4.5.2.2.1 Individual tests may continue. For production reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of indicators on hand or produced later shall not be made until it is determined that all indicators meet all requirements of the specification.

4.5.3 Defects in indicators already accepted. The investigation of a test failure could indicate that defects may exist in indicators already accepted. If so, the contractor shall fully advise the qualifying activity of all defects likely to be found and the method of correcting them (see 6.2).

4.6 Test methods

4.6.1 Examination of product. The indicator shall be inspected to determine compliance with the requirements specified herein with respect to material, workmanship, marking, and dimensions.

4.6.2 Dielectric strength. After sealing, 250V at a commercial frequency shall be applied between those pins not connected to ground and the case for a period of 5 seconds. There shall be no insulation breakdown. All lighting pins shall be considered grounded for this test.

4.6.3 Scale error and friction

4.6.3.1 Scale error. The indicator shall be connected to a master transmitter. The transmitter shall be set at angles equivalent to each of the test points as specified in table III. The indicator shall read within the tolerances specified therein. The output of the indicator power supply shall also be within the tolerances specified in 3.5.3.1 and table I.

4.6.3.2 Friction. The indicator shall be tested for friction at the scale mark nearest to 5, 10, 50, and 90 percent of full scale. The master transmitter shall be made to slowly approach the test point in an ascending direction. After the transmitter has been held at the test point for approximately 5 seconds, the indicator pointer reading shall be recorded. The indicator shall then be gently vibrated to establish a stable condition and the pointer reading shall be recorded. The difference between the two readings shall be recorded as friction and shall not exceed the tolerances specified in table I.

4.6.4 Hysteresis. The master transmitter shall be made to slowly approach the test point nearest to the 20, 50, and 90 percent of full-scale test point and the indicator shall be tapped. Readings shall be taken at the test points from both up-scale and down-scale directions. The difference in the readings shall not exceed the tolerances specified in table I.

4.6.5 Response. The indicator shall respond and indicate a half-scale or full-scale reading within 1 percent when a half-scale step input is applied from zero to half-scale and from half-scale to full-scale, respectively, in the following time periods at the specified temperatures. The indicator shall be energized for at least 3 minutes prior to this test.

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Temperature (°C)	Indicator (Second Max)
-54	10
+25	3
+71	3

4.6.6 Sealing. The indicator case shall be tested for leaks by means of a mass-spectrometer-type helium leak detector. The detected leak rate shall not exceed 0.1 micron cubic foot per hour at a pressure differential of 1 atmosphere.

* 4.6.7 Voltage and frequency variation. This test shall be conducted with the indicator connected to a load impedence as shown on figure 5. Power shall be applied to obtain the following combinations of voltage and frequency: 104 ±1V 320 ±2 Hz, 126 ±1V 360 ±2 Hz, 117 ±1V 320 ±2 Hz, 104 ±1V 440 ±2 Hz, 113 ±1V 480 ±2 Hz, 126 ±1V 480 ±2 Hz. For each voltage-frequency combination, the accuracy of the indications shall be determined by positioning the master transmitter to obtain simulated flow rates of 5, 10, 50, and 90 percent of full scale. The accuracy of the indicator flow rates and the output of the regulated power supply shall be as specified in table I. The error of the master transmitter shall be determined at the specified test points and with the specified voltages and frequencies prior to testing. The transmitter error shall be algebraically subtracted from the indicated flow rates to determine the indicated error.

4.6.8 Frequency cycling. With the indicator connected as specified in 4.6.7, the voltage shall be adjusted to 115 ±1V and the frequency cycled from 320 to 480 Hz in accordance with the following schedule with approximately half-scale flow.

a. Start cycle at 320 Hz, increase frequency to 480 Hz in 38 seconds lineally with time.

b. Return frequency to 320 Hz in 2 seconds or less.

c. Increase frequency to 480 Hz in 58 seconds lineally with time and return to 320 Hz within 2 seconds.

This 100-second cycle shall be repeated continuously for 10 hours at the rate of 36 cycles per hour. Power shall then be removed for a period of 2 ±1/2 hours and shall be reapplied at 121 ±1V 320 ±2 Hz for 1 hour. Immediately following this 1-hour period, the indicator shall be frequency cycled for 1 hour in accordance with the schedule indicated in a, b, and c except the frequency range shall be between 300 and 500 Hz. The indicator shall then be subjected to the scale error test at room temperature.

4.6.9 Voltage surge. The indicator shall be connected as shown on figure 5. This test shall be made at approximately 5, 50, and 90 percent of full scale. Rated power shall be applied for 30 minutes. The power input shall then be adjusted to 108 ±1V 480 ±2 Hz for 1 hour and adjusted immediately to 121 ±1V 480 ±2 Hz for 30 minutes. At the end of this 2-hour operating period, the power input shall be adjusted to 115 ±1V 320 ±2 Hz. A voltage surge shall then be applied to provide an rms peak voltage of 160 ±5V and return to 115 ±1V. The duration of the surge, including rise and decay time, shall be 0.09 ±0.005 second. The frequency shall then be increased to 480 ±2 Hz and a voltage surge shall be applied to provide a momentary rms voltage of 195 ±5V and return to 115 ±1V. The duration of the surge, including rise and decay time, shall be 0.07 ±0.005 second. At the end of this test, the voltage and frequency shall

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be adjusted to 115 ±1V 400 ±2 Hz and a scale error test at room temperature shall be conducted. The errors shall not exceed the room temperature tolerances of table III.

* 4.6.10 Vibration error. The indicator shall be subjected to vibration test in accordance with MIL-STD-810, Method 514.2, category b.2, except procedure I, table 514.2-II, figure 514.2-2, curve B shall be used for one 15 minute sweep cycle of 5-500-5 Hz in each of the three mutually perpendicular planes. The indicator shall be mounted in its normal position and shall be energized to provide an indication of approximately half-scale. Pointer oscillation and variation shall not exceed the tolerances specified in table I.

4.6.11 Low temperature. The indicator, not energized, shall be subjected to a temperature of -54° ±2°C for a period of 4 hours. While at this temperature, the master transmitter shall be set at zero pph. Power shall be applied to the indicator. The output of the power supply shall be monitored for compliance with 3.5.3.2 and table II. At the end of a 10-minute period, the master transmitter shall be set to the half-scale test point, and the time for the indicator to reach this point shall not exceed 25 seconds. The indicator shall then be subjected to the scale error test and shall meet the scale error tolerances specified in table III. At the conclusion of this test and while the indicator is still in a -54°C ambient, the response-time test shall be conducted. This test shall be conducted immediately after the low temperature scale error test and prior to application of power to the indicator for an accumulation of 40 minutes.

4.6.12 High temperature. The indicator, while energized, shall be subjected to a temperature of 71° ±2°C for a period of 4 hours, and tested at this temperature for scale errors. The scale errors shall not exceed the tolerances specified in table III. At the conclusion of this test and while the indicator is still in a 71°C ambient, the response test shall be conducted. This test shall be conducted immediately after the high temperature scale error test. The output of the power supply shall be checked for compliance with 3.5.3.2 and table II.

4.6.13 High altitude-low temperature. The indicator shall be subjected to an absolute pressure equivalent to an altitude of approximately 100,000 feet for a period of 1 hour at a temperature of -55° ±2°C. While still at this pressure and temperature, the indicator shall be subjected to the scale error test specified in 4.6.3. The scale error shall not exceed the tolerances specified in table III. There shall be no damage to the indicator as a result of this test.

4.6.14 Position error. The scale error test shall be repeated at the scale mark nearest to 5, 10, 50, and 90 percent of full scale with the instrument turned 90° right, 90° left, and face down. The change in scale error shall not exceed the tolerances specified in table I.

4.6.15 Damping. The input signal to the indicator shall be made to oscillate about the scale mark nearest to the 5, 10, 50, and 90 percent of full scale test points with a double amplitude of 3 percent of full scale and at a frequency between 2.0 and 2.4 Hz. The indicator pointer shall oscillate with a total excursion less than 2 percent of full scale.

4.6.16 Thermal shock and fogging. The indicator shall be subjected to a total of 8 cycles of immersion in tap water at a temperature of 85° ±2°C and then in tap water at a temperature of 5° ±2°C. The length of time for each bath immersion shall be 30 minutes with not more than 3 seconds between immersions.

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During the fourth and eighth cycles at the end of the 30-minute soak in tap water at 85°C, the indicator shall be removed from the hot bath and within 4 seconds, the face of the instrument shall be inserted in the 5°C tap water to a depth of 0.75 ±0.25 inch. At the end of 1 minute, the indicator shall be withdrawn from the 5°C bath and the cover glass immediately observed for internal fogging. If no fogging is observed, the sequency of the test shall be continued to completion. No evidence of moisture penetration or damage to the enclosure shall result. The indicator shall then be resubjected to and shall pass the test specified in 4.6.6. Fogging of the cover glass during any part of this test shall be cause for rejection.

4.6.17 Power consumption. The indicator shall be connected with a load impedance as shown on figure 5 and power shall be applied. The power consumption shall not exceed 30 va, exclusive of lighting power and power consumed by the master transmitter.

* 4.6.18 Low temperature exposure. The low temperature exposure test shall be conducted in accordance with MIL-STD-810, Method 502.1. The indicator shall be subjected to a temperature of -62°C for a period of 48 hours after which the temperature shall be raised to -54°C and maintained for 24 hours. At the end of the 24-hour period and while still at -54°C, the indicator shall be subjected to the scale error and response tests. The scale error shall not exceed the tolerance of 4.6.11 and the response time shall be as specified in 4.6.5 for -54°C. The temperature shall be raised to room temperature and maintained for not less than 4 hours after which the indicator shall be subjected to the scale error test at room temperature and then the response test at room temperature. The scale error shall not exceed the tolerances specified in table III and the response time shall not exceed that specified in 4.6.5 for room temperature.

* 4.6.19 High temperature exposure. The high temperature exposure test shall be conducted in accordance with MIL-STD-810, Method 501.1. After the 48-hour period and while at 71°C, the indicator shall be tested for scale error. The errors shall not exceed those specified in 4.6.12. The indicator shall then be maintained at room temperature for a period of not less than 4 hours after which it shall be subjected to the room temperaure scale error test specified herein. The errors observed shall not exceed the tolerance specified in table III. The indicator shall be examined for deterioration or damage due to the extreme high temperature exposure.

4.6.20 Magnetic effect. The indicator, not operating, shall be rotated in a vertical plane about a short bar magnet compass with the nearest part of the indicator 5.50 inches from the magnetically east or west of the center of the compass. Starting directly under the compass, the indicator shall be held in positions, 0°, 45°, 135°, 180°, 225°, 270°, and 315° from the initial position. At each of these positions, the indicator shall be rotated on its own horizontal axis until it is in the normal upright position. The horizontal magnetic field intensity shall be 0.17 and 0.19 oersted. The deflection of the compass at any of the specified position shall not exceed 3°. The test shall be repeated with the indicator operating at rated voltage.

* 4.6.21 Salt fog. The indicator shall be subjected to a salt fog test in accordance with MIL-STD-810, Method 509.1, procedure I. The scale error at room temperature test shall be conducted immediately before and after this test. The scale error tolerance after the salt fog test shall not exceed the room temperature tolerances specified in table III. Subsequent indicator operation shall not be affected by this test, and excessive corrosion of metal parts, including electrical receptacle, shall be cause for rejection.

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4.6.22 Electromagnetic interference. The electromagnetic interference test shall be conducted in accordance with MIL-STD-462 except that the frequency range shall be from 0.15 to 150 mc. Test methods shall be as follows: CE02, CE03, CE04, CS01, CS02, CS06, RS01, RS02, and RS03.

4.6.23 Fogging. The indicator shall be energized and kept in a 70° ±2°C controlled ambient temperature for a minimum of 1 hour. At the end of this period and while still at the high temperature, an ice cube shall be rubbed on the face of the indicator glass for a period of 1 to 2 minutes. The glass shall then be wiped dry (compressed air shall not be used) and the indicator inspected for evidence of water and oil fog. Any evidence of fogging shall be cause for rejection. This test may be conducted in conjunction with the high temperature scale error and friction tests. (See 4.6.12 and 4.6.3.2.)

* 4.6.24 Lighting tests (brightness and color)

* 4.6.24.1 Lighting. The following tests shall be performed as individual tests during qualification and sampling tests.

* 4.6.24.1.1 Light. The indicator shall be subjected to the light tests (brightness, color, and stray light), as specified in MIL-L-25467 and MIL-L-27160, as applicable.

* 4.6.24.1.2 Subarea measurements. All areas including the indicator areas specified in 3.9 shall be measured.

* 4.6.24.2 Comparison tests. The following tests shall be conducted only during the individual production quality conformance testing:

* 4.6.24.2.1 Visual comparison. This test shall be conducted by qualified personnel only after they have been tested for normal color vision and have spent at least 20 minutes in the dark to become dark adapted. The prime standard and the indicator under test shall be placed side-by-side at eye level in completely dark surroundings and viewed at a distance of approximately 2.5 feet in a sighting cone 45° from the normal to the surface of the cover glass. All indicator markings, pointers, et cetera, shall be compared when viewed from any position within the sighting cone.

* 4.6.24.2.2 Brightness. With 2.50 ±0.01V applied to the indicator under test and 2.40 and 2.60V applied consecutively to the lighting terminal of the standard, like areas of the pointers, numerals, indicia, markings, et cetera of the test indicator and the prime standard shall be compared for brightness. The two instruments shall also be compared for lighting uniformity. The brightness of the production unit at 2.50V shall fall within the brightness levels of the prime standard when set at 2.40 and 2.60V. If the brightness is dissimilar or exceeds the limits of the prime standard in any individual area or the relative uniformity of lighting of the indicator and prime standard are different, the questionable areas of the production units shall be measured with a photometer in accordance with MIL-L-25467 or MIL-L-27160, as applicable. If the production units fail this test, they shall be rejected and returned for rework. Qualification units shall not be tested by use of the prime standard. The qualification units must be photometrically measured per the applicable lighting specification.

* 4.6.24.2.3 Color. The color comparison test shall be conducted as specified in 4.6.24.1.1, except 4.50V shall be applied to both the production indicator and the prime standard. The two indicators shall appear identical in color.

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4.6.25 Compatibility. Compatibility test shall be conducted as required to demonstrate compliance with 3.5.1, 3.5.2, 3.5.3 and 3.5.7. Transmitter starting tests shall also be conducted on the EFU-15A/A series indicators. The indicator shall be properly connected to the appropriate TRU-31A/A transmitter which has been filled with fuel and capped off. A total of 100 starting cycles shall be made with 104 ±1V 320 Hz power applied to the indicator. The transmitter shall not fail to start each time power is applied.

* 4.6.26 Vibration endurance. The indicator shall be subjected to a vibration test in accordance with MIL-STD-810, Method 514.2, category b.2, procedure I, table 514.2-II, figure 514.2-2, curve B. The indicator shall be operating and shall be vibrated in each of the three mutually perpendicular planes with a signal input of approximately half scale. Following the vibration period, the indicator shall be subjected to the scale error, friction error and hysteresis tests. Room temperature tolerances shall be within those specified, and there shall be no evidence of loosened or damaged parts as a result of the applied vibration.

* 4.6.27 Shock. The indicator shall be subjected to a shock test in accordance with MIL-STD-810, Method 516.2, procedure I, except the shock force shall be 15g. Immediately after this test, the indicator shall be subjected to the room temperature scale error test. The error observed shall not exceed the tolerance specified for the applicable indicator.

4.6.28 Reliability

4.6.28.1 Qualification. The 11 indicators specified in 4.4.1 shall be subjected to a reliability test in accordance with MIL-STD-781, test level E, which shall demonstrate a minimum acceptable MTBF of 1,200 hours to a 90 percent confidence level.

4.6.28.1.1 Test equipment. The test equipment and fixtures shall include provisions for indicator operation cycling, temperature cycling, and vibration.

4.6.28.1.2 Operation cycling. During this test, the indicator shall be continuously cycled from 5 to 90 percent of full scale and return at a rate of one cycle every 2 to 4 minutes.

4.6.28.1.3 Temperature cycling. Unless otherwise specified or amended by the results of the thermal analysis, the temperature of the chamber in which the indicators are tested shall be cycled as follows:

a. Starting at room temperature, decrease temperature to -54°C and stabilize for 1-1/2 hours with indicators unenergized.

b. Turn indicators on, raise temperature to +55°C, and operate indicators for 4 hours.

c. Turn indicators off and lower chamber temperature to -54°C for 2 hours.

d. Repeat steps b and c until reliability requirement has been met. This schedule provides four 6-hour test cycles for each 24-hour period. In each 24-hour period, the operational test time is 16 hours.

4.6.28.1.4 Vibration cycling. Throughout this test, the indicators shall be vibrated for a period of 10 minutes during each hour of on-time at any nonresonant frequency between 20 and 60 Hz and at a peak acceleration of 2.2g ±10 percent.

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4.6.28.1.5 Performance check. At each 200-hour interval and at the end of this test, the indicators shall be subjected to the following: Scale error, friction, hysteresis, response time, damping, and lighting (color and brightness tests). During each scale error test, if any two tests points exceed the allowable tolerances specified in table I or III, as applicable, one failure shall be charged against this test. The lighting circuit shall be energized during this entire test. The indicators shall meet the lighting tolerances of the applicable lighting specification for the first 1,000 hours of testing on each indicator.

4.6.28.1.6 Failure criteria. For an accept decision, each indicator shall be operated at least 700 hours. If an indicator fails, entries shall be made on the appropriate data logs and the failed equipment shall be removed from the test and repaired with minimum interruption of the indicators still under test. The qualifying activity shall be immediately notified of the failure and the plan for correcting the deficiency. A detailed analysis of the failure will be submitted as soon as it has been completed. No more than a total of 12 failures will be permitted during this test. Acceptance shall be based on the accept-reject criteria specified in MIL-STD-781, test plan I, after each indicator has been operated for at least 700 hours.

4.6.28.1.7 Reporting. Reports shall be submitted in accordance with MIL-STD-781. The reliability test log and the failure summary and analysis shall be included in the final test report.

* 4.6.28.2 Quality conformance. The sample indicators specified in 4.5.2.1 shall be subjected to a reliability test in accordance with the "Reduced Testing" requirements of MIL-STD-781, test level E, and as specified under 4.6.28.1, which shall demonstrate a minimum acceptable MTBF of 1,200 hours to a 90 percent confidence level. Accept-reject criteria shall be in accordance with test plan V except the discrimination ratio shall be 1.5. When Tightened Testing is required by the qualifying activity, acceptance shall be in accordance with test plan I or III as determined by that activity. Results of this test shall be included in the test report specified in 4.6.28.1.7.

4.6.28.3 Longevity. Testing of 3 of the 11 reliability test samples shall be continued through a longevity test in accordance with MIL-STD-781, test plan XXVIII, until 2,500 hours of operation have accumulated on each indicator.

4.7 Inspection of preservation, packaging, packing, and marking for shipment and storage. Sample items or packs and the inspection of preservation, packaging, packing and marking for shipment and storage shall be in accordance with section 5 or the documents specified therein.

5. PACKAGING

5.1 Preservation-packaging. Preservation and packing shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Indicators shall be preserved and packaged Method III in accordance with MIL-P-116 and MIL-STD-794, one each in a unit container conforming to PPP-B-636, domestic class.

5.1.2 Level C. Indicators shall be individually preserved and packaged in a manner that will afford adequate protection against corrosion, deterioration, or physical damage during shipment from the supply source to the first receiving activity.

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5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Indicators preserved and packaged as specified in 5.1.1 shall be packed in weather-resistant shipping containers conforming to PPP-B-636. Shipments that exceed the weight limitations of this specification shall be packed in overseas-type containers conforming to PPP-B-601. The gross weight of each PPP-B-601 container shall not exceed 200 pounds. Where practicable, exterior shipping containers shall be of uniform shape and size and of minimum cube and tare. Closure and strapping shall be in accordance with the appendix of PPP-B-636 or PPP-B-601, as applicable. Waterproofing shall be in accordance with MIL-STD-1186.

5.2.2 Level B. Indicators preserved and packaged as specified in 5.1.1 shall be packed in domestic shipping containers conforming to PPP-B-636. Where practicable, exterior shipping containers shall be of uniform shape and size and of minimum cube and tare. The gross weight of each shipping container shall not exceed the weight limitations of the specification. Closure and strapping shall be in accordance with the appendix of PPP-B-636.

5.2.3 Level C. Indicators preserved and packaged as specified in 5.1.2 shall be packed adequately to assure carrier acceptance and safe delivery at the lowest rate to the initial receiving activity. All containers shall conform to commercial freight handlers rules and regulations as applicable to the mode of transportation.

5.3 Marking for shipment. Interior and exterior containers shall be marked for shipment in accordance with MIL-STD-129. The nomenclature shall be as follows:

Indicator Rate of Flow EFU-15A/A- ____*____ or EFU-42/A- ____*____
____**____ PPH, 2-Inch Size ____***____

* Applicable type designation
** Applicable range
*** Applicable lighting.

5.3.1 Reinspection date markings. The reinspection date markings shall be as specified by the procuring activity.

6. NOTES

6.1 Intended use. The EFU-15A/A indicators covered by this specification are intended for use in displaying rate of fuel flow in pounds per hour being consumed by the aircraft engine. The indicators also supply power to the transmitter motor. The EFU-42/A indicators do not incorporate a power supply and are used as a repeater display.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Type of indicator required (see 1.2).

c. That the indicators specified in 4.5.3 be returned to the contractor for appropriate corrective action at the contractor's expense.

d. Levels of packaging and packing required and reinspection date markings (see section 5).

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6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order than they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Aeronautical Systems Division, Attn: ENAID, Wright-Patterson Air Force Base, Ohio 45433 and information pertaining to qualification of products may be obtained from that activity.

6.3.1 The qualification test procedures will be submitted for formal approval before the test program is initiated.

6.3.2 Test samples. The qualification test samples will be submitted to the qualifying activity.

6.4 Identification of changes. The margins of this specification are marked with an asterisk to indicate where changes (additions, modificatins, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force - 11

Review activity:
Air Force - 99

Preparing activity:
Air Force - 11

Project No. 6620-F114

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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